LAB # 8

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Lab Section : C

1. Introduction

The objective of this experiment was to design, build, and test an AM radio receiver by combining several basic electronic circuit blocks studied throughout the course. The receiver was designed to process an amplitude-modulated signal around 175 kHz and recover the original audio-frequency message signal. To achieve this, the circuit was divided into smaller stages: a tuning circuit, an amplification stage, an envelope detector, and a final audio amplification stage. Instead of building the whole receiver at once, each block was implemented and tested step by step to verify its operation before connecting it to the next stage.

Amplitude modulation is a communication technique in which a low-frequency message signal is carried by a high-frequency carrier signal. In this lab, the carrier frequency was around 175 kHz, while the message signal represented the audio information. Since the received AM signal is usually very small, it must first be selected by a tuner and then amplified before detection. For frequency selection, a parallel LC circuit was used as the tuner. The resonant frequency of this circuit is determined by the values of the inductor and capacitor, and it allows signals near the desired frequency to pass while attenuating signals at other frequencies.

After the tuner stage, the signal was amplified using cascaded op-amp amplifier stages. In the pre-lab design, each amplifier stage was designed to have a gain of approximately 5, giving an overall first-stage gain of about 125 when three stages were cascaded. This approach provides enough amplification while maintaining better bandwidth performance than using one very high-gain stage. The amplified AM signal was then applied to the detector stage. The detector circuit, consisting of a diode, resistor, and capacitor, performs envelope detection by following the outer shape of the AM waveform and extracting the original message signal. Finally, the recovered audio signal was amplified again so that it could drive the speaker or headphones.

This experiment demonstrates how RLC circuits, diodes, capacitors, and op-amps can be combined to form a complete communication system. It also shows the importance of testing each circuit block separately, because the performance of the final receiver depends on the correct operation of every stage. Through this lab, the practical roles of tuning, amplification, demodulation, and audio output in an AM radio receiver were observed and analyzed.

2. Pre-Lab Work

In this pre-lab, the aim is to design the main parts of an AM radio receiver before implementing the full system in the lab. The receiver will include a tuning circuit to select the desired frequency and amplifier stages to increase the signal level. The target frequency for this lab is 175 kHz, so the circuit should be designed accordingly.

The first part of the design is the LC tuning circuit. For a parallel LC circuit, the resonance frequency is given by $f_o = 1 / (2\pi\sqrt{LC})$. Here, the inductance is given as L = 8.2 µH and the desired resonance frequency is 175 kHz. Rearranging the formula to solve for the capacitance: $C = 1 / ((2\pi f_o)^2 L)$

After substituting the values, the capacitance is found to be approximately 100 nF. Therefore, the tuning circuit will be implemented using L = 8.2 µH and C = 100 nF in parallel. This circuit will allow signals around 175 kHz to pass while attenuating other frequencies.

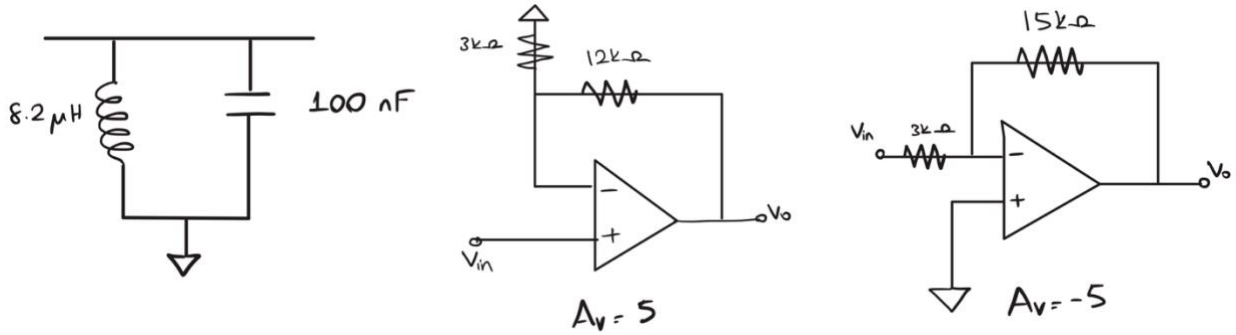


Figure 1. Schematic of LC and op-amp circuits.

The second part is designing the amplifier stages. According to the lab requirements, one non-inverting and one inverting amplifier should be designed, each with a gain of 5, and all resistor values must be greater than 1 kΩ. These amplifier stages will later be cascaded to obtain sufficient overall gain for the received signal.

For the non-inverting amplifier, the voltage gain is given by $A_v = 1 + Rf / R1$. To obtain a gain of 5, the ratio Rf / R1 must be equal to 4. A suitable choice is R1 = 3 kΩ and Rf = 12 kΩ, which satisfies the required gain: $A_v = 1 + 12k / 3k = 5$

For the inverting amplifier, the voltage gain is $A_v = -Rf / Rin$. To achieve a gain of -5, the ratio Rf / Rin should be 5. Choosing Rin = 3 kΩ and Rf = 15 kΩ gives $A_v = -15k / 3k = -5$

In the complete system, one non-inverting and two inverting amplifiers will be connected in cascade. This results in an overall gain magnitude of $|A_{v,total}| = 5 \times 5 \times 5 = 125$. Such a high gain is necessary because the signal received from the antenna is very small and needs to be amplified before it can be processed by the detector stage.

3. Data/Results

Part-1: Tuner Stage

Building Block: LC Tuner / Band-Pass Filter

In the first part of the experiment, the tuner stage of the AM radio receiver was built and tested. The purpose of this stage was to select the desired carrier frequency around 175 kHz and attenuate signals at other frequencies before the amplification stage. In an AM receiver, the tuner is a very important block because many signals may be present at the input, but only the desired frequency should be passed to the rest of the circuit. For this reason, a parallel LC circuit was used as a band-pass filter.

Components:
8.2 µH inductor
100 nF capacitor
Breadboard
Function generator
Oscilloscope
Jumper wires and probe connections

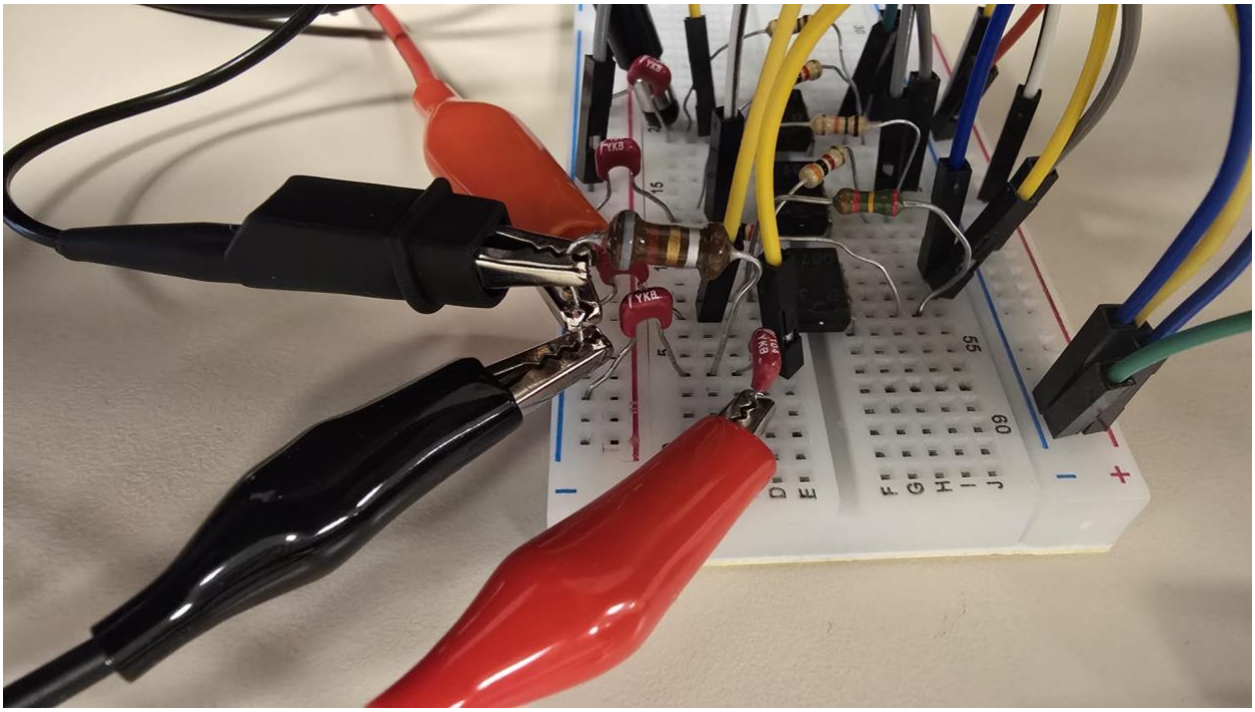


Figure 2. Breadboard implementation of the LC tuner stage.

The tuner circuit was implemented using an 8.2 µH inductor and a 100 nF capacitor connected as an LC resonant circuit. These values were selected from the resonance equation $f_o = 1 / (2\pi\sqrt{LC})$

Using L = 8.2 µH and C = 100 nF, the theoretical resonant frequency becomes approximately $f_o \approx 175$ kHz

This means that the circuit is expected to respond most strongly to signals around 175 kHz, which is the target carrier frequency of the AM receiver.

During the experiment, the function generator was connected to the input of the tuner stage and the oscilloscope was used to observe the output signal. The input signal amplitude was adjusted to 1 V, and the frequency was set near the target resonance region. According to the oscilloscope measurement, the output waveform was observed at a frequency of approximately 175.78 kHz. The measured output peak-to-peak voltage was about 0.82 V.

Measurement:

Input voltage: 1 V, Output voltage: 0.82 V, Resonance frequency: 175.78 kHz
A clear sinusoidal waveform was obtained at the output, which shows that the tuner stage was operating properly near its resonant frequency. Since this block is a band-pass filter and not an amplifier, the main purpose was not to increase the signal amplitude, but to allow the desired frequency component to pass while suppressing unwanted frequency components. Therefore, obtaining an output voltage of 0.82 V for a 1 V input is reasonable and expected.



Figure 3. Oscilloscope output of the tuner stage showing an output of approximately 0.82 V at 175 kHz.

The fact that the output is lower than the input can be explained by several practical factors. First, real inductors and capacitors are not ideal and have parasitic resistances and losses. Second, the breadboard introduces additional parasitic effects. Third, the oscilloscope and source connections load the circuit and reduce the measured output amplitude. Because of these non-ideal conditions, the band-pass filter does not pass the signal with perfect unity gain. Instead, it passes the desired frequency with some attenuation, which is still sufficient to demonstrate correct tuning behavior.

This result confirms that the LC tuner successfully worked as a band-pass filter centered near 175 kHz. The measured frequency was very close to the theoretical resonance frequency, and the output waveform showed that the desired signal component could pass through the tuner stage. Therefore, the first part of the AM receiver was successfully implemented and verified.

Part-2: Amplification Stage

Building Block: Three-Stage Op-Amp Amplifier

In the second part of the experiment, the first amplification stage of the AM receiver was implemented and tested. The purpose of this block was to increase the amplitude of the tuned signal before applying it to the detector stage. Since the signal received by the antenna and passed by the tuner is usually small, amplification is necessary for the detector circuit to recover the message signal properly.

The amplification stage was built using TL081 operational amplifiers. The complete amplifier block consisted of three cascaded op-amp stages. The first stage was designed as a non-inverting amplifier, and the following two stages were designed as inverting amplifiers. Each stage was designed to have a voltage gain magnitude of approximately 5. Therefore, the expected total voltage gain of the cascaded amplifier was $A_{v,total} = 5 \times 5 \times 5 = 125$

Components:
TL081 op-amps
3 kΩ resistors
12 kΩ resistor
15 kΩ resistors
100 nF capacitor for noise suppression
Breadboard
Function generator
Oscilloscope
DC power supply
Jumper wires and probe connections

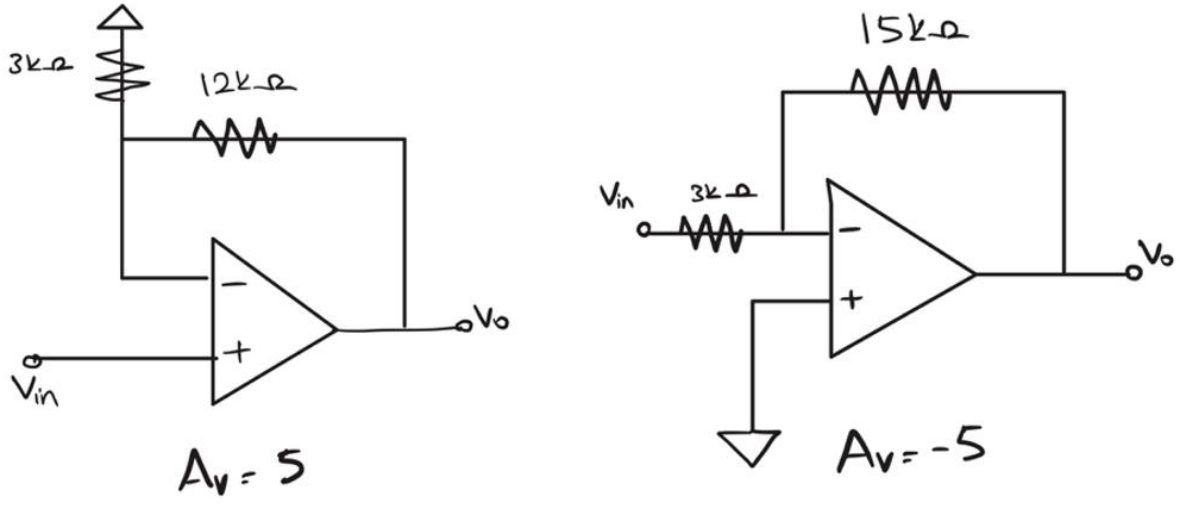


Figure 4. Schematic of the non-inverting and inverting amplifier stages.

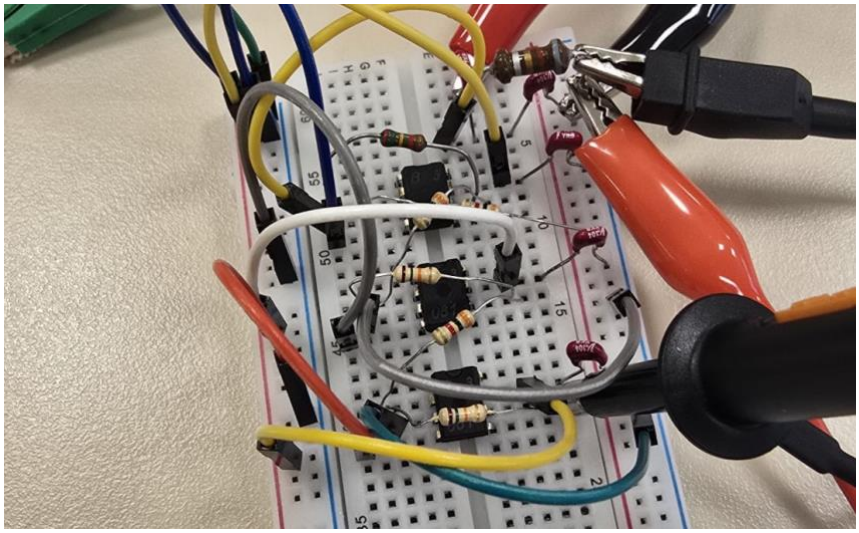


Figure 5. Breadboard implementation of the TL081 cascaded amplifier stage.

For the non-inverting amplifier, 3 kΩ and 12 kΩ resistors were used. The gain of a non-inverting amplifier is given by Av = 1 + Rf / R1. Substituting the selected resistor values: Av = 1 + 12 kΩ / 3 kΩ = 1 + 4 = 5 Therefore, the first stage amplified the signal by a factor of 5 without changing the phase of the signal.

For each inverting amplifier stage, 3 kΩ was used as the input resistor and 15 kΩ was used as the feedback resistor. The gain of an inverting amplifier is Av = -Rf / Rin Using the selected values: Av = -15 kΩ / 3 kΩ = -5

Each inverting stage therefore provided a gain magnitude of 5 and inverted the signal phase by 180°. Since two inverting stages were used after the non-inverting amplifier, the signal was inverted twice. As a result, the final output of the three-stage amplifier had an overall positive gain with respect to the input.

The theoretical total gain was calculated as Av,total = 5 × (-5) × (-5) = 125. Thus, for a 100 mVpp input signal, the ideal expected output would be Vout,ideal = 100 mVpp × 125 = 12.5 Vpp

During the experiment, a 100 mVpp sine wave was applied from the function generator. The circuit output was measured using the oscilloscope. Around the desired operating frequency, the output waveform was clean and sinusoidal. The maximum measured output was approximately 9.7 Vpp at 175.78 kHz.

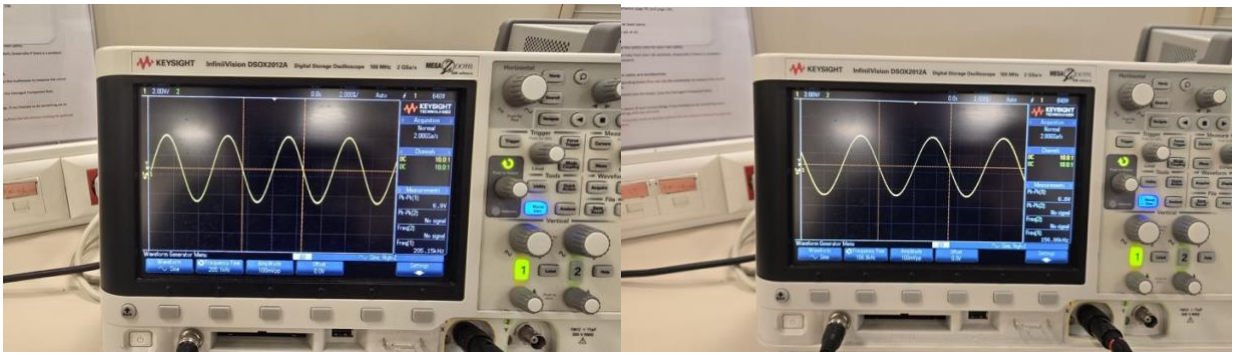


Figure 6. Oscilloscope outputs near the lower and upper 3 dB frequencies, showing approximately 6.8 Vpp at 156.96 kHz and 6.9 Vpp at 205.15 kHz.

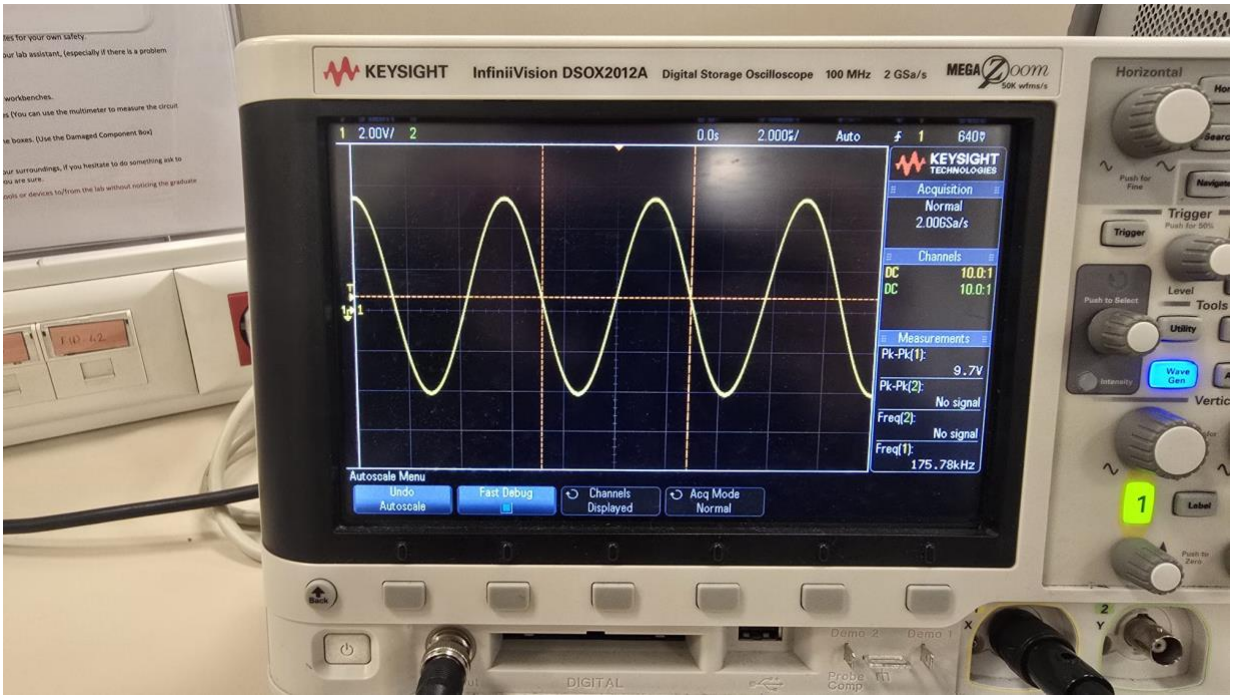


Figure 7. Oscilloscope output at 175.78 kHz showing approximately 9.7 Vpp output for 100 mVpp input.

Measurement:

Input voltage: 100 mVpp
Maximum output voltage: 9.7 Vpp
Measured frequency at maximum output: 175.78 kHz
Experimental gain:
Av,exp = Vout / Vin
Av,exp = 9.7 V / 0.1 V = 97

The measured gain was approximately 97 V/V. This value is lower than the theoretical gain of 125 V/V, but it is still reasonably close for a practical cascaded op-amp circuit. The difference between theoretical and experimental gain can be explained by non-ideal op-amp behavior, resistor tolerances, breadboard parasitic effects, loading between stages, and the bandwidth limitation of the TL081 op-amp at high frequency. In theory, the gain equations assume ideal op-amps, but in practice the op-amp gain decreases as frequency increases because of the gain-bandwidth limitation.
After verifying the gain, a frequency versus magnitude test was performed. The output voltage was measured at different frequencies around the operating region. The maximum output was 9.7 Vpp, so the 3 dB output level was calculated as
$V_{3dB} = V_{max} / \sqrt{2}$
$V_{3dB} = 9.7\ V / \sqrt{2} \approx 6.86\ V$
The oscilloscope measurements showed that the output decreased to approximately this level at both sides of the peak frequency.
Frequency response measurements:
Input Voltage | Frequency | Output Voltage | Gain
100 mVpp | 156.96 kHz | 6.8 Vpp | 68 V/V
100 mVpp | 175.78 kHz | 9.7 Vpp | 97 V/V
100 mVpp | 205.15 kHz | 6.9 Vpp | 69 V/V

The lower 3 dB frequency was approximately 156.96 kHz, and the upper 3 dB frequency was approximately 205.15 kHz. Therefore, the bandwidth of the measured response was
Bandwidth = fH - fL
Bandwidth = 205.15 kHz - 156.96 kHz = 48.19 kHz
This result shows that the amplifier stage provided the strongest output near the desired AM carrier frequency. At 175.78 kHz, the output reached its maximum measured value of 9.7 Vpp. When the frequency was moved away from this region, the output voltage decreased to around 6.8–6.9 Vpp, which corresponds closely to the 3 dB points. This confirms that the receiver chain produced the highest response around the intended operating frequency.
The 100 nF capacitor was included between the supply and ground to reduce noise and stabilize the op-amp operation. This is important because high-gain cascaded amplifier circuits can easily amplify unwanted noise or oscillations. By adding the decoupling capacitor, sudden supply variations and high-frequency noise were reduced, making the output waveform more stable.

Part-3: Detector Stage

Building Block: Envelope Detector / AM Demodulator

In the third part of the experiment, the detector stage of the AM radio receiver was implemented and tested. The main purpose of this stage was to recover the original low-frequency message signal from the amplified AM signal. In the previous parts of the experiment, the tuner stage selected the desired carrier frequency around 175 kHz, and the amplification stage increased the amplitude of this selected signal. However, even after amplification, the signal was still an amplitude-modulated high-frequency signal. This means that the useful audio information was not directly the carrier waveform itself, but the slowly varying envelope of that waveform. Therefore, a detector stage was required to remove the carrier component and extract the message signal.

Components:
1N4148 diode
22 nF capacitor
10 kΩ resistor
Breadboard
Function generator
Oscilloscope
Jumper wires and probe connections
Output signal from the previous amplification stage

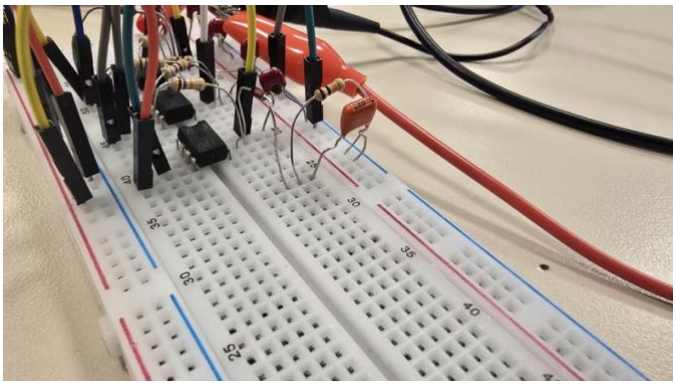


Figure 8. Breadboard implementation of the detector stage using a diode, 22 nF capacitor, and 10 kΩ resistor.

The detector circuit was built using a diode, a 22 nF capacitor, and a 10 kΩ resistor, as specified in the lab manual. This stage is also called an envelope detector because it follows the envelope of the AM signal and produces an output that represents the original message signal. The lab manual describes this detector stage as the part where the AM signal is demodulated and the audio signal is pulled out from the radio signal.

The AM signal used in this experiment can be expressed as: $s(t) = [1 + k\cos(\omega_m t)] A \cos(\omega_c t)$

In this expression, $A \cos(\omega_c t)$ represents the carrier signal, and $k\cos(\omega_m t)$ represents the message signal. The carrier frequency is much higher than the message frequency. In this experiment, the carrier frequency was approximately 174–175 kHz, while the message frequency was approximately 1 kHz. The reason for using a high-frequency carrier is that low-frequency audio signals cannot be efficiently transmitted or received directly. Instead, the message signal is placed onto the amplitude of a high-frequency carrier. Then, at the receiver side, the detector circuit removes the carrier and recovers the original message signal.

The envelope detector works by using the charging and discharging behavior of the capacitor. When the positive half-cycle of the AM signal reaches a peak, the diode becomes forward-biased and allows current to flow. During this time, the capacitor charges quickly to approximately the peak value of the input signal. After the peak passes, the input voltage begins to decrease. At this point, the capacitor voltage becomes higher than the instantaneous input voltage, so the diode becomes reverse-biased and stops conducting. Then, the capacitor discharges through the 10 kΩ resistor. This process repeats continuously for each carrier cycle.

Because the capacitor charges rapidly at the peaks and discharges more slowly through the resistor, the output voltage follows the outer shape of the AM waveform rather than following every high-frequency oscillation of the carrier. This outer shape is the envelope, and it corresponds to the original message signal. Therefore, the detector converts the high-frequency AM waveform into a lower-frequency signal that can later be amplified and applied to a speaker or headphones.

The resistor and capacitor values are very important for successful envelope detection. The RC time constant determines how fast the capacitor discharges: $\tau = RC$

For this circuit: $\tau = (10\ k\Omega)(22\ nF)$ $\tau = 220\ \mu s$

This time constant should be selected carefully. If the time constant is too small, the capacitor discharges too quickly between the carrier peaks. In that case, the output contains too much ripple and cannot produce a smooth envelope. If the time constant is too large, the capacitor discharges too slowly and cannot follow the changes in the message signal. This causes distortion because the output becomes delayed and does not accurately track the envelope.

In this experiment, the carrier frequency was around 174 kHz. The carrier period is approximately: $T_c = 1 / f_c = 1 / 174000 \approx 5.75\ \mu s$. The message frequency was approximately 1 kHz. The message period is: $T_m = 1 / f_m = 1 / 1000 = 1\ ms$

The selected time constant, 220 µs, is much larger than the carrier period but smaller than the message period. This is a suitable condition for envelope detection because the capacitor does not discharge

significantly during each very short carrier cycle, while it is still able to follow the slower 1 kHz envelope. Therefore, the 10 kΩ resistor and 22 nF capacitor are appropriate values for this detector stage.

Normally, a Germanium diode is preferred instead of a regular silicon diode because it has a lower forward voltage drop. This is important in detector circuits since the signal amplitude may not be very large. A lower forward voltage drop allows the diode to conduct more easily and helps the capacitor charge even when the input signal amplitude is relatively small. If a silicon diode with a higher forward voltage drop is used, part of the signal may be lost before detection, resulting in a lower output amplitude. However, since a Germanium diode was not available in the lab, a regular silicon diode was used in this experiment.

During the experiment, the detector stage was connected after the amplification stage. First, the AM signal before detection was observed on the oscilloscope. This signal still contained the high-frequency carrier, and its amplitude varied according to the message signal. The oscilloscope showed a carrier frequency of approximately 174.67 kHz and a peak-to-peak voltage of approximately 4.9 Vpp. This confirmed that the signal entering the detector was an amplified AM waveform with the expected carrier frequency.

After the detector circuit was connected, the output waveform was measured across the resistor-capacitor part of the circuit. The oscilloscope showed that the output signal had a frequency of approximately 999.9 Hz. This value is very close to the expected 1 kHz message frequency. The measured peak-to-peak output voltage was approximately 940 mVpp, and the DC RMS value was approximately 676 mV.

Measurement:

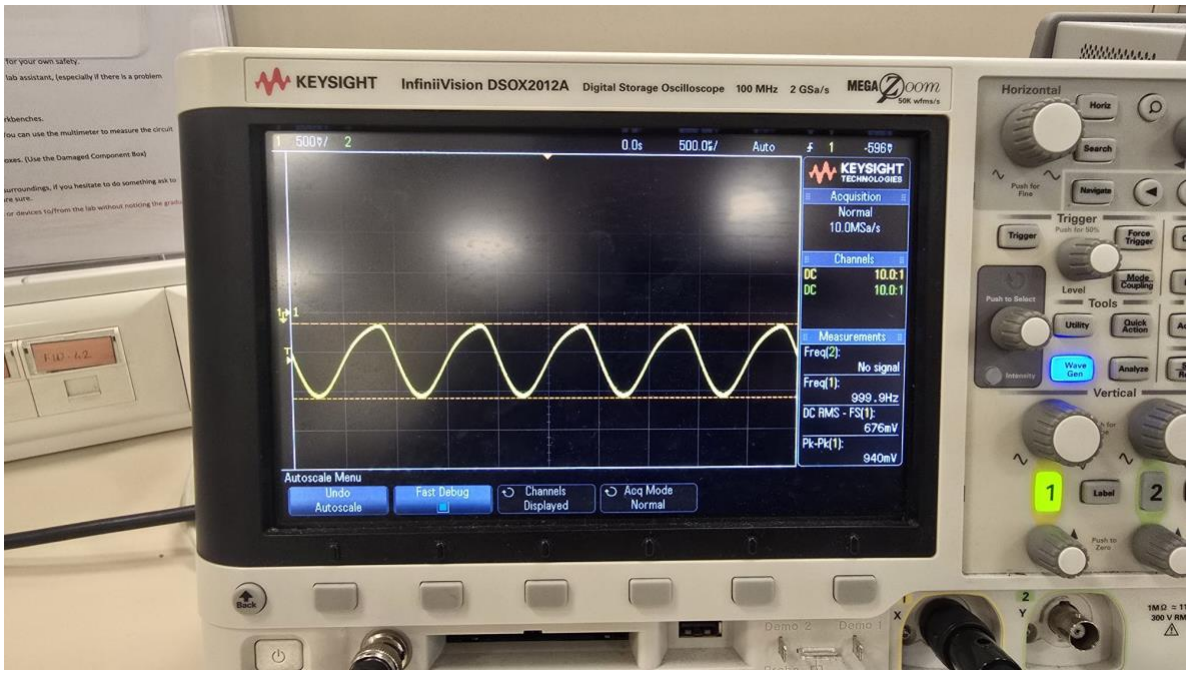


Figure 9. Detector output waveform showing the recovered message signal at approximately 999.9 Hz with 940 mVpp output amplitude.

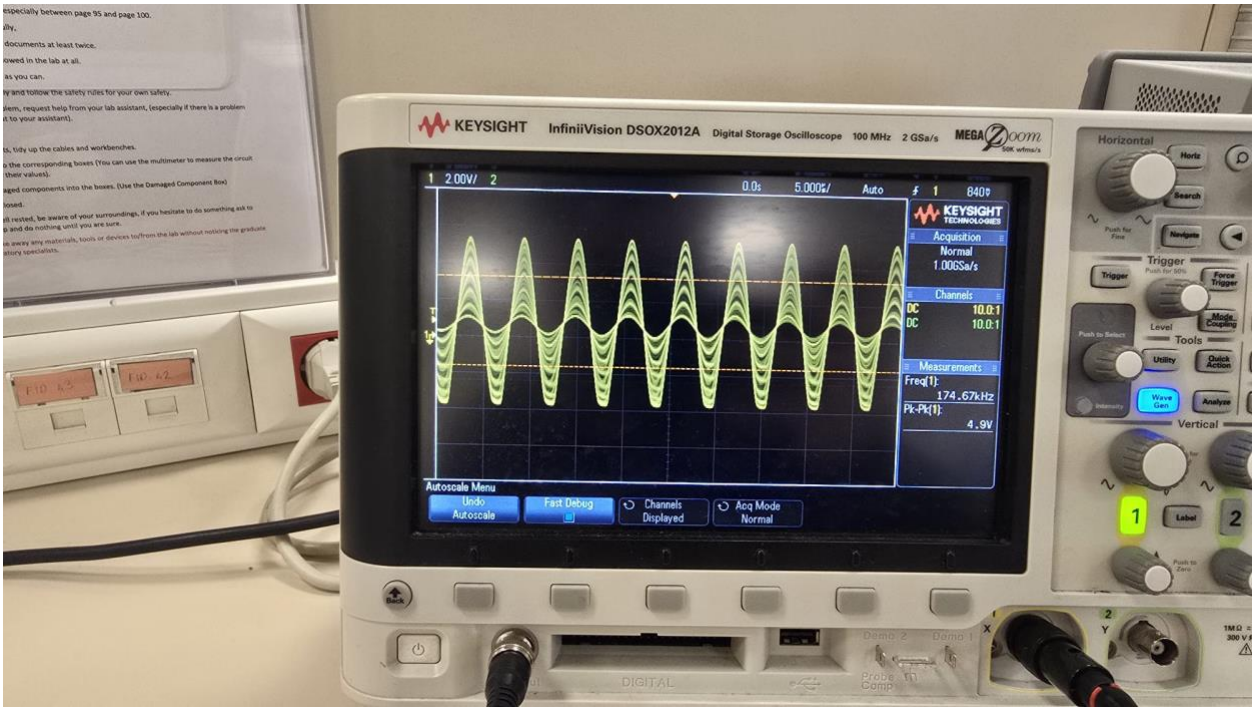


Figure 10. AM signal before detection, showing approximately 174.67 kHz carrier frequency and 4.9 Vpp amplitude.

The AM signal before detection had a carrier frequency of 174.67 kHz, while the detected output signal had a recovered frequency of 999.9 Hz with a peak-to-peak voltage of approximately 940 mVpp.

These measurements show that the detector stage worked properly. Before the detector, the signal was a high-frequency AM waveform around 174–175 kHz. After the detector, the output frequency became approximately 1 kHz, which means that the message signal was successfully recovered. The output signal was not expected to have the same frequency as the carrier because the purpose of the detector is not to preserve the carrier, but to extract the envelope of the carrier.

The detected output was also lower in amplitude than the AM signal before detection. This is expected because the detector is not an amplifier. It only extracts the envelope and removes most of the carrier component. Some voltage is also lost due to the diode forward voltage drop and the charging-discharging behavior of the capacitor. In addition, real circuit effects such as component tolerances, breadboard parasitics, oscilloscope loading, and imperfect connections can reduce the measured output amplitude.

The output waveform was close to the expected detected signal shape, but it was not perfectly smooth. This is also normal for a practical envelope detector. A small ripple remains because the capacitor discharges slightly between consecutive carrier peaks. If the capacitor were larger, the ripple would decrease, but the output might not follow the message signal quickly enough. If the capacitor were smaller, the circuit would follow the envelope more quickly, but the ripple would increase. Therefore, the selected RC value represents a balance between reducing carrier ripple and preserving the shape of the 1 kHz message signal.

he detector stage successfully demodulated the AM signal and recovered the original low-frequency message signal. The measured output frequency of 999.9 Hz was very close to the expected 1 kHz message frequency, which confirms that the envelope detection process worked correctly. This stage is essential in the AM receiver because it converts the amplified high-frequency carrier waveform into an audio-frequency signal that can be processed by the final amplification stage and used to drive the speaker or headphones.

Part-4: Final Audio Amplification Stage

Building Block: Audio Amplifier Stage

In the fourth part of the experiment, the final audio amplification stage of the AM receiver was implemented and tested. After the detector stage, the original message signal was recovered from the AM waveform, but the detected signal level was still not high enough to directly drive the output load effectively. Therefore, a final amplifier stage was required to increase the amplitude of the recovered audio-frequency signal before sending it to the speaker or output terminal.

Components:

TL081 op-amp
120 kΩ feedback resistor
10 kΩ gain resistor
100 kΩ bias resistor
100 µF output coupling capacitor
Breadboard
Oscilloscope
Function generator / AM signal input
DC power supply
Jumper wires and probe connections

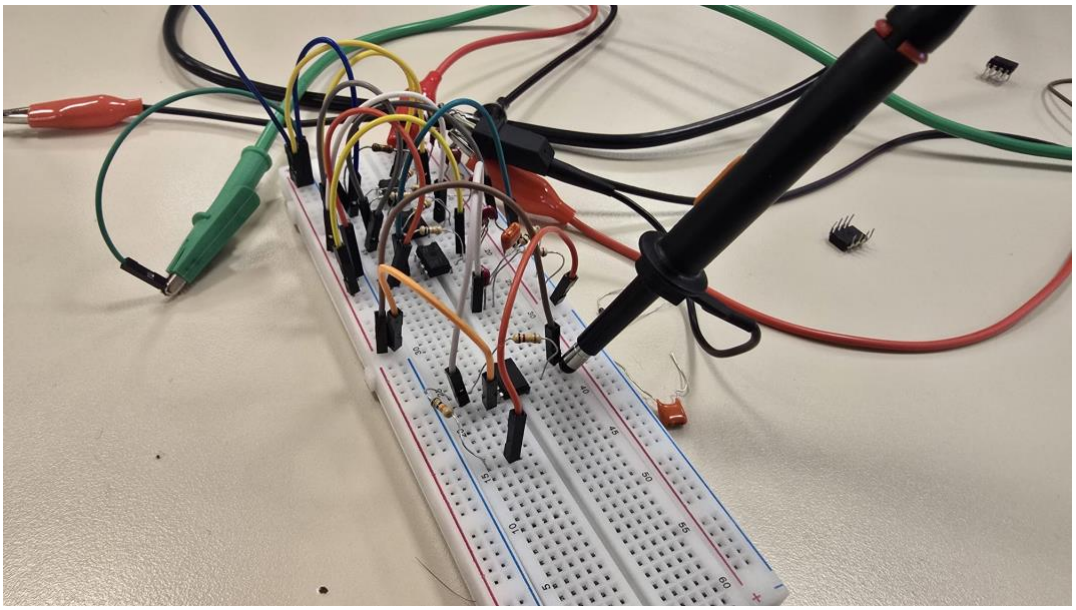


Figure 11. Breadboard implementation of the final audio amplification stage.

The audio amplifier stage was built after the detector output. In this part, the detected 1 kHz message signal was applied to the input of the final amplifier. The amplifier was designed using a 120 kΩ resistor and a 10 kΩ resistor for the gain-setting network. The gain of this stage can be estimated using the non-inverting amplifier gain equation:

Av = 1 + Rgain / Rgain2

Using the selected resistor values: Av = 1 + 120 kΩ / 10 kΩ Av = 13

Therefore, the final stage was expected to amplify the detected audio signal by approximately 13 times. This high gain was useful because the detector output was less than 1 Vpp in the previous stage, and the signal needed to be increased before the output stage. The 100 kΩ resistor was used to provide a proper reference/bias path at the amplifier input, while the 100 µF capacitor was used as the output coupling capacitor. This capacitor blocks the DC component of the amplifier output and allows mainly the AC audio signal to pass to the speaker or load. This is important because a DC component across the speaker would not contribute to sound and could also cause unnecessary power dissipation.

In this stage, the signal being amplified was no longer the high-frequency carrier signal. Instead, it was the recovered audio-frequency message signal from the detector stage. For this reason, the oscilloscope output was expected to show a waveform around 1 kHz. This was confirmed experimentally. The measured frequency at the output was approximately 1.0009 kHz, which matches the expected audio/message frequency.

During the experiment, the modulation percentage was also observed to affect the output amplitude. When the AM signal was tested with 100% modulation, the oscilloscope showed an output of approximately 26.5 Vpp at the final amplifier output. This means that the final stage significantly increased the detected signal amplitude and produced a strong audio-frequency output.

Measurements

Measurement at 100% modulation: Recovered output frequency: 1.0009 kHz
Output voltage: 26.5 Vpp

When the modulation depth was reduced to approximately 30%, the final output voltage also decreased. In this case, the measured output was approximately 8 Vpp. This result is expected because the modulation index determines how strongly the carrier amplitude changes according to the message signal. A higher modulation percentage creates a larger envelope variation, so the detector produces a larger recovered message signal. Since the final amplifier amplifies this detected message signal, a larger modulation depth results in a larger final output voltage. Similarly, when the modulation depth is reduced, the envelope variation becomes smaller, and the final amplified output also becomes smaller.

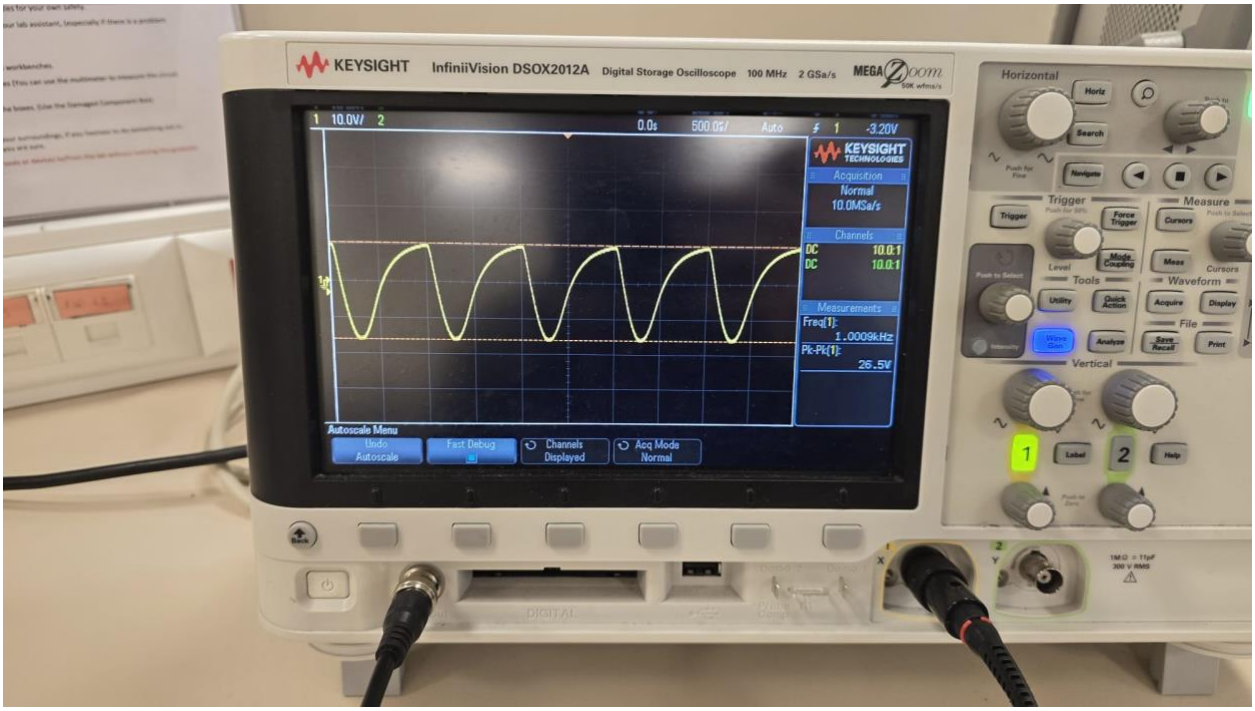


Figure 12. Oscilloscope output of the final audio amplifier at 100% modulation, showing approximately 1.0009 kHz frequency and 26.5 Vpp output.

Measurement at 30% modulation: Output voltage: approximately 8 Vpp

The difference between the 100% modulation and 30% modulation outputs shows that the receiver responds directly to the depth of amplitude modulation. At 100% modulation, the envelope of the AM signal has a larger variation, so the detector output is stronger and the final amplifier output reaches about 26.5 Vpp. At 30% modulation, the envelope variation is smaller, so the final output is reduced to around 8 Vpp. This behavior confirms that the detector and final amplifier stages are working together correctly.

The shape of the output waveform was close to the expected recovered audio signal, but it was not perfectly sinusoidal. Some distortion and waveform asymmetry can be observed, especially at high output amplitude. This is reasonable because the amplifier was producing a large output voltage, and practical op-amps have output swing limitations depending on the supply voltage. At very large amplitudes, the output can approach the supply limits, which may cause slight clipping or distortion. In addition, the detector output itself is not a perfectly pure sine wave because it contains ripple from the envelope detection process. When this non-ideal detected signal is amplified, the imperfections also become more visible at the final output.

Part-5: Complete AM Receiver Test and Listening

Building Block: Complete AM Radio Receiver System

In the final part of the experiment, all previously tested stages were connected together to form the complete AM radio receiver. The aim of this part was to verify that the full circuit could operate as one system, starting from receiving or applying an AM signal and ending with an audible output from the speaker. At this point, the tuner, amplification stage, detector stage, and final audio amplifier had already been tested separately. Therefore, this part focused on checking the combined operation of the entire receiver chain.

Components:

Antenna setup
LC tuner stage
Three-stage op-amp amplification block
Diode-capacitor detector stage
Final audio amplifier stage
Speaker
Function generator
DC power supply
Breadboard
Jumper wires and probe connections

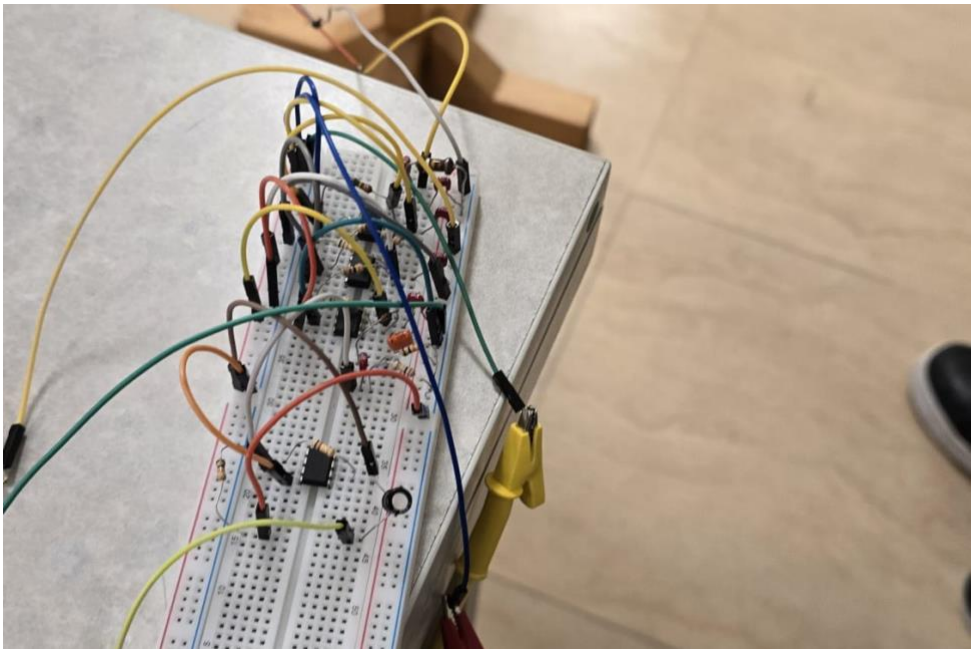


Figure 13. Breadboard implementation of the complete AM receiver circuit.

The complete receiver was tested step by step. First, an AM modulated sine wave was applied from the function generator to verify that the receiver circuit could process a known input signal. This was important because using the function generator made it easier to control the carrier frequency,

modulation depth, and signal amplitude. After confirming that the output waveform was proper, the circuit was connected to the antenna setup under the guidance of the TA.

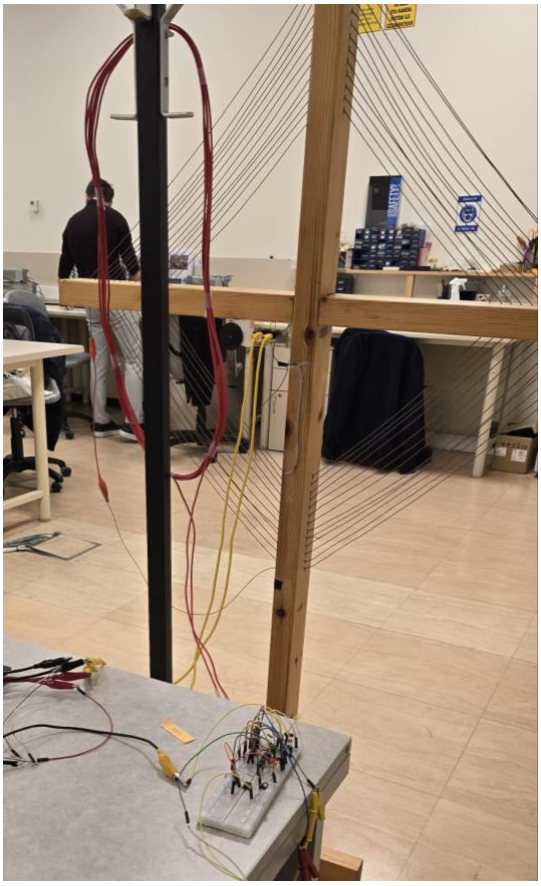


Figure 14. Complete AM receiver setup connected to the antenna and speaker.

The overall operation of the receiver can be explained as a chain of functional blocks. First, the antenna receives the AM signal from the environment. Then, the LC tuner selects the desired carrier frequency around 175 kHz and reduces unwanted frequency components. After this, the first amplification stage increases the amplitude of the tuned AM signal so that it is strong enough for detection. The detector stage then removes most of the carrier component and recovers the low-frequency message signal from the envelope of the AM waveform. Finally, the audio amplifier increases the recovered signal level and drives the speaker.

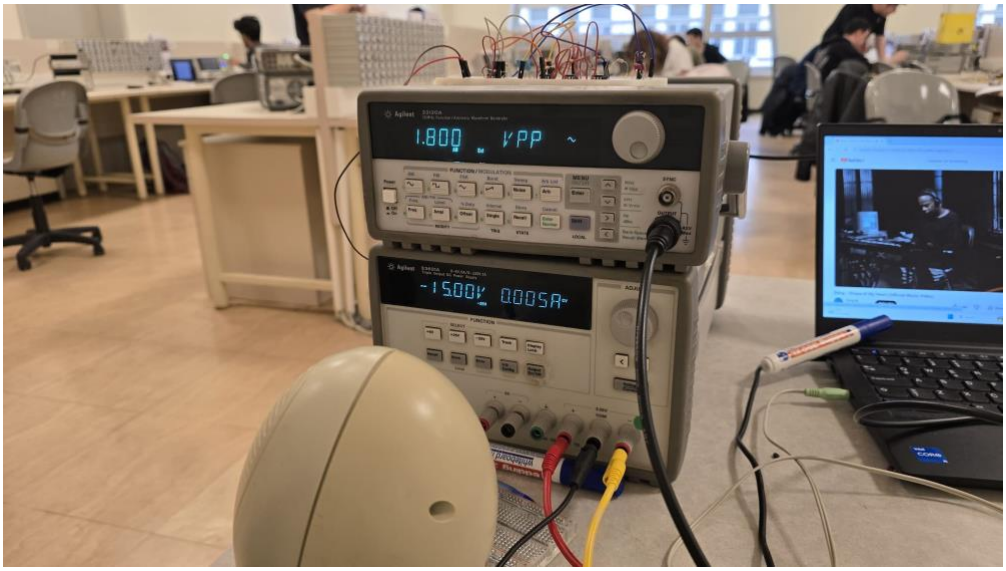


Figure 15. Function generator and power supply used during the complete receiver test.

During the final test, the circuit worked successfully as a complete AM receiver. The signal passed through each stage properly, and the recovered audio signal could be heard from the speaker. This showed that the tuner, amplifier, detector, and final audio amplifier were all functioning together correctly. The sound output confirmed that the envelope detection and final amplification stages were able to convert the AM signal into an audible signal.

The circuit performance was generally good. Since all stages were built on a breadboard with many jumper wires, some noise and small distortions were expected. These effects can come from loose connections, breadboard parasitic capacitance, long wires, component tolerances, and environmental noise picked up by the antenna. However, despite these practical limitations, the circuit produced a clear enough output, and the receiver operation was successfully demonstrated.

4. Discussion/Analysis

In this experiment, a complete AM radio receiver was designed, implemented, and tested by dividing the system into smaller functional blocks. The receiver consisted of a tuner stage, a three-stage op-amp amplification block, an envelope detector, a final audio amplifier, and finally the antenna-speaker test. This step-by-step approach was important because the complete receiver was a relatively large circuit, and testing each block separately made it easier to find possible problems before connecting the whole system together. The main purpose of the receiver was to select an AM carrier signal around 175 kHz, amplify it, recover the original audio-frequency message signal, and then amplify this recovered signal enough to drive the speaker.

The general signal flow of the receiver was consistent with the expected AM receiver operation. First, the LC tuner selected the desired carrier frequency. Then, the first amplification stage increased the amplitude of the selected AM signal. After that, the detector stage extracted the envelope of the AM signal, which corresponds to the original message signal. Finally, the audio amplifier increased the recovered message signal so that it could be heard from the speaker. Since the final circuit produced a clear audible output, the receiver was considered successful.

A summary of the main theoretical and measured results is given below:

Stage	**Theoretical / Expected Result**	**Measured Result**	**Comment**
LC tuner	Resonance near 175 kHz	175.78 kHz	Very close to expected value
LC tuner output	Should pass 175 kHz signal	0.82 V output for 1 V input	Signal passed with some attenuation
First amplifier block	Gain = 5 × 5 × 5 = 125	Gain ≈ 97	Lower than ideal but still strong amplification
Amplifier max output	12.5 Vpp for 100 mVpp input ideally	9.7 Vpp	Difference due to non-ideal op-amp behavior and losses
Amplifier 3 dB points	Output should drop to about 0.707 of maximum	6.8 Vpp at 156.96 kHz and 6.9 Vpp at 205.15 kHz	Consistent with 3 dB definition
Detector output	Recover 1 kHz message	999.9 Hz, 940 mVpp	Message signal successfully recovered
Final audio amplifier	Gain ≈ 13	26.5 Vpp at 100% modulation, about 8 Vpp at 30% modulation	Output increased as modulation depth increased
Complete receiver	Audible output expected	Speaker output was obtained	Full receiver worked successfully

Table 1.

The first stage was the LC tuner. In theory, the resonant frequency of a parallel LC circuit is found from $f_o = 1 / (2\pi\sqrt{LC})$

Using L = 8.2 µH and C = 100 nF, the expected resonant frequency is approximately 175 kHz. In the experiment, the measured frequency was 175.78 kHz, which is very close to the theoretical value. This agreement shows that the selected inductor and capacitor values were appropriate for tuning the AM carrier frequency. The input voltage was 1 V, and the measured output was 0.82 V. This means the circuit did not amplify the signal, but it successfully passed the desired frequency with some attenuation. This is expected because the tuner is a passive band-pass filter, not an amplifier. In a passive LC circuit, energy

losses occur due to the internal resistance of the inductor, capacitor tolerance, breadboard parasitics, oscilloscope loading, and source resistance. Therefore, an output slightly lower than the input is reasonable.

The LC tuner is important because an antenna can pick up signals at many different frequencies. If the circuit did not have a tuning stage, unwanted frequency components could also be amplified and detected. The tuner makes the receiver frequency-selective. In this experiment, since the measured frequency was almost exactly at the target value, the tuner successfully prepared the correct signal for the next stage.

The second stage was the three-stage TL081 op-amp amplifier. The first amplifier was designed as a non-inverting amplifier using 12 kΩ and 3 kΩ resistors. Its theoretical gain was $A_v = 1 + 12\ k\Omega / 3\ k\Omega = 5$

The next two amplifiers were inverting amplifiers, each using 15 kΩ and 3 kΩ resistors. Their gains were $A_v = -15\ k\Omega / 3\ k\Omega = -5$

Since the two inverting stages invert the signal twice, the total gain becomes positive: $A_{v,total} = 5 \times (-5) \times (-5) = 125$

For a 100 mVpp input, the ideal output would therefore be 12.5 Vpp. Experimentally, the maximum output was measured as 9.7 Vpp, giving an experimental gain of $A_{v,exp} = 9.7\ V / 0.1\ V = 97$

The measured gain was lower than the theoretical gain. This difference is expected because the ideal gain equations assume ideal op-amps, perfect resistors, no loading between stages, and unlimited bandwidth. In the real circuit, the TL081 does not behave ideally at all frequencies. Also, the resistor values have tolerances, the breadboard adds parasitic capacitances and resistances, and long jumper wires can introduce noise and additional loading. Since three op-amp stages were cascaded, small losses or phase shifts in each stage could accumulate and reduce the final output amplitude.

Another important point is the gain-bandwidth product of the op-amp. In the lab, three amplifier stages with moderate gain were used instead of one amplifier with a very large gain. This is a good design choice because increasing the closed-loop gain of a single op-amp reduces its available bandwidth. If a single op-amp stage were designed with a gain such as 64 or 125, its bandwidth would become much smaller, and it would not amplify the 175 kHz carrier properly. By distributing the total gain across three stages, each stage only needed a gain of 5, so each stage could keep a wider bandwidth. This allowed the overall amplifier block to provide high gain while still operating around the carrier frequency.

The frequency response of the amplifier stage also supported this behavior. The maximum output was 9.7 Vpp, so the 3 dB level was approximately $V_{3dB} = 9.7 / \sqrt{2} \approx 6.86$ Vpp

Experimentally, the output was about 6.8 Vpp at 156.96 kHz and about 6.9 Vpp at 205.15 kHz. These values are very close to the expected 3 dB level. Therefore, the lower and upper 3 dB frequencies were approximately 156.96 kHz and 205.15 kHz. The bandwidth was $BW = 205.15\ kHz - 156.96\ kHz = 48.19\ kHz$

This bandwidth was sufficient for the AM receiver because the signal around 175 kHz could pass and be amplified effectively. The maximum response near the desired carrier frequency confirmed that the amplifier block was working properly with the tuner stage.

The third stage was the envelope detector. This stage was necessary because the output of the amplifier was still an AM waveform, not the original audio signal. In amplitude modulation, the high-frequency carrier carries the message by changing its amplitude according to the lower-frequency signal. The detector extracts this changing envelope. The detector circuit used a diode, a 22 nF capacitor, and a 10 kΩ resistor. Ideally, a Germanium diode is preferred because of its lower forward voltage drop, but in this experiment a regular silicon diode was used because a Germanium diode was not available.

The detector operates by charging the capacitor during the peaks of the AM waveform and then allowing it to discharge through the resistor between peaks. The RC time constant was $\tau = RC = (10\ k\Omega)(22\ nF) = 220\ \mu s$

This value is suitable because it is much larger than the carrier period and smaller than the message period. For a carrier frequency around 174 kHz, the carrier period is about 5.75 µs. For a 1 kHz message signal, the message period is 1 ms. Since 220 µs is between these two time scales, the capacitor does not discharge too much during each carrier cycle, but it can still follow the 1 kHz envelope. This is why the detector was able to recover the message signal.

Experimentally, the AM signal before detection had a carrier frequency of 174.67 kHz. After the detector, the recovered output signal had a frequency of 999.9 Hz and a peak-to-peak voltage of approximately 940 mVpp. This result shows that the detector successfully removed most of the high-frequency carrier and recovered the original message signal. The output was not a perfect sine wave, which is expected. The use of a silicon diode caused a larger voltage drop than a Germanium diode would have caused, and the capacitor discharge created some ripple. However, the detected signal frequency was almost exactly 1 kHz, so the envelope detection process worked correctly.

The fourth stage was the final audio amplifier. After detection, the recovered signal had a much lower frequency and lower amplitude than the amplified carrier signal. This signal needed to be amplified again before being applied to the speaker. In this stage, 120 kΩ and 10 kΩ resistors were used, giving an approximate non-inverting gain of $A_v = 1 + 120\ k\Omega / 10\ k\Omega = 13$

At 100% modulation, the final output was measured as approximately 26.5 Vpp at about 1 kHz. At 30% modulation, the output was about 8 Vpp. This result is consistent with AM theory. When the modulation percentage increases, the envelope variation becomes larger. Since the detector output follows the envelope, a higher modulation depth produces a larger recovered signal. Then, the final amplifier increases this signal further. Therefore, the 100% modulation case produced a much larger output than the 30% modulation case.

Some distortion was visible at the final output, especially at the larger 26.5 Vpp output level. This is also reasonable because the op-amp was working with a large output swing. When the output amplitude becomes large, the op-amp may approach its supply voltage limits, causing clipping or waveform distortion. In addition, the detector output already contains some ripple and non-ideal shape, and the final amplifier amplifies these imperfections along with the useful signal. Even so, the measured frequency was correct and the output amplitude was large enough for the speaker stage.

In the final part, the complete receiver was connected and tested with the antenna and speaker. This was the most important system-level test because it showed whether all blocks could work together. The antenna received the AM signal, the tuner selected the desired frequency, the amplifier increased the

carrier amplitude, the detector recovered the audio signal, and the final amplifier drove the speaker. The circuit worked well, and the sound was successfully heard from the speaker. This confirmed that the complete receiver chain was functioning correctly.

There were still practical limitations in the final circuit. The circuit was built on a breadboard with many jumper wires, so noise pickup and parasitic effects were unavoidable. The antenna connection could also introduce environmental noise. The speaker output level depended on modulation depth, amplifier gain, power supply limits, and the quality of the detector output. Also, using a silicon diode instead of a Germanium diode likely reduced the detector efficiency. However, these limitations did not prevent the receiver from operating. The circuit produced the expected output behavior at each stage and worked successfully as a full AM radio receiver.

In general, the experimental results were consistent with the theoretical expectations. The tuner passed the desired carrier frequency near 175 kHz, the amplifier increased the signal level significantly, the detector recovered the 1 kHz message signal, and the final amplifier produced a strong audio output. The measured values were not exactly equal to the ideal calculations, but the differences were reasonable for a real breadboard implementation. Component tolerances, non-ideal op-amp behavior, gain-bandwidth limitations, diode voltage drop, loading effects, and parasitic capacitances all contributed to these differences. Despite these non-idealities, the circuit performed its intended function successfully.

This experiment showed how basic circuit components can be combined to create a complete communication system. The LC circuit provided frequency selection, the op-amps provided gain, the diode-capacitor network performed demodulation, and the final amplifier made the signal usable for a speaker. Each block had a specific role, and the receiver only worked properly when all of them were connected in the correct order. The successful final output demonstrated that the design choices made in the pre-lab and in-lab stages were appropriate for building a working AM radio receiver.

5. Post-Lab

Question 1. Explain why we used three stages with gain of 5 in order to obtain a gain of $5 \times 5 \times 5 = 125$ given the fact that gain bandwidth product of the op-amp that we used in the lab is 4 MHz. Why did we not use a single op-amp with a gain of 64? Explain in detail.

We used three cascaded amplifier stages with a gain of 5 each because of the gain-bandwidth product limitation of the op-amp. In an op-amp, when the closed-loop gain increases, the available bandwidth decreases. The approximate relation is: GBW = Gain × Bandwidth

In this lab, the gain-bandwidth product of the op-amp was 4 MHz. If we used a single op-amp with a gain of 64, the bandwidth would be approximately: Bandwidth = 4 MHz / 64 = 62.5 kHz
This bandwidth is too low for our AM receiver because the carrier frequency that we needed to amplify was around 175 kHz. If the amplifier bandwidth is only about 62.5 kHz, the 175 kHz signal would be strongly attenuated. Therefore, the amplifier would not be able to amplify the AM carrier properly.

If we tried to obtain the total gain of 125 using only one op-amp stage, the bandwidth would be even smaller: Bandwidth = 4 MHz / 125 = 32 kHz

This would be much worse, because 32 kHz is far below the 175 kHz carrier frequency. The signal would be heavily reduced before reaching the detector stage, and the envelope detector would not receive a strong enough AM waveform to demodulate.

Instead, we divided the total gain into three smaller stages. Each stage had a gain of 5, so the bandwidth of each stage was approximately: Bandwidth = 4 MHz / 5 = 800 kHz

This bandwidth is much higher than 175 kHz, so each amplifier stage can pass and amplify the AM carrier signal effectively. When the three stages are cascaded, the total gain becomes: $A_{v,total} = 5 \times 5 \times 5 = 125$

This design gives a high total gain while keeping the bandwidth of each individual stage large enough. In practical circuit design, it is usually better to use several moderate-gain stages instead of one very high-gain stage when both high gain and high bandwidth are required. This also reduces distortion, improves stability, and prevents the op-amp from operating too close to its frequency limit.

In our experiment, the theoretical total gain was 125, but the measured gain was approximately 97 because 100 mVpp input produced about 9.7 Vpp output. The difference is reasonable because real op-amps are not ideal. The gain-bandwidth limitation, resistor tolerances, loading between stages, breadboard parasitics, and high-frequency behavior of the TL081 all reduce the practical gain. However, the three-stage design still provided enough amplification for the detector stage and worked better than a single high-gain amplifier would have.

Question 2. Which type of filter do you use to pass 175 kHz frequency component? How can a multichannel receiver/radio be designed and what changes to the filter are required?
To pass the 175 kHz frequency component, we used an LC band-pass filter as the tuner stage. The tuner was built using an 8.2 µH inductor and a 100 nF capacitor. This LC circuit was designed to resonate around 175 kHz. At resonance, the circuit allows the desired carrier frequency to pass more effectively while attenuating frequencies far from the resonant frequency.

The resonant frequency of the LC circuit is given by: $f_0 = 1 / (2\pi\sqrt{LC})$
Using L = 8.2 µH and C = 100 nF, the theoretical resonant frequency is approximately 175 kHz. Experimentally, we measured the output at approximately 175.78 kHz, which is very close to the expected value. This shows that the filter worked properly as a tuner stage.

The reason we need this filter is that an antenna can receive many different frequency components from the environment. Without a frequency-selective filter, unwanted signals could enter the amplifier and detector stages. The band-pass filter selects the desired carrier frequency before amplification, so the receiver processes mainly the intended AM signal.

A multichannel receiver or radio can be designed by making the tuning frequency adjustable. Since the resonant frequency depends on L and C, the receiver can select different channels by changing either the capacitance or the inductance. In real radios, this is commonly done using a variable capacitor or a varactor diode. When the capacitance changes, the resonant frequency of the LC circuit shifts, allowing the receiver to tune to different carrier frequencies.

For example, if the capacitor value is decreased, the resonant frequency increases. If the capacitor value is increased, the resonant frequency decreases. Therefore, a multichannel receiver would require a tunable band-pass filter instead of a fixed LC filter. The filter should also have an appropriate bandwidth. If the bandwidth is too wide, multiple channels may pass at the same time and interfere with each other. If the bandwidth is too narrow, part of the desired AM signal may be distorted or attenuated. Therefore, the quality factor, or Q factor, of the filter must be selected carefully.

In real-life radio receiver design, the tuner stage is very important because it determines selectivity. A good receiver should select the desired channel clearly while rejecting nearby channels and noise. This is why practical radios use adjustable and carefully designed band-pass filters.

Question 3. What is the function of diode and capacitor in detecting the signal envelope? Explain how the detector/demodulator works.
The diode and capacitor are the main components of the envelope detector. Their function is to extract the low-frequency message signal from the high-frequency AM carrier signal. In an AM signal, the information is stored in the envelope of the carrier. The detector stage follows this envelope and removes most of the high-frequency carrier component.

The AM signal can be written as: $s(t) = [1 + k\cos(\omega m\, t)]\, A \cos(\omega c\, t)$
Here, $\cos(\omega c\, t)$ is the carrier signal and $\cos(\omega m\, t)$ is the message signal. In this lab, the carrier frequency was around 174–175 kHz, while the message frequency was approximately 1 kHz. The detector needed to recover the 1 kHz message from the 175 kHz carrier.

The diode works like a rectifier. During the positive peaks of the AM signal, the diode becomes forward-biased and allows current to flow. This charges the capacitor quickly to approximately the peak value of the AM waveform. When the input signal decreases after the peak, the diode becomes reverse-biased and stops conducting. Then, the capacitor discharges slowly through the resistor.

This repeated charging and discharging makes the output follow the envelope of the AM signal. The capacitor does not follow every high-frequency carrier cycle. Instead, it smooths the waveform and follows the slower changes in amplitude. These slower changes correspond to the original message signal.

The RC time constant is very important for correct detection: $\tau = RC$
In this experiment, we used: $R = 10\ k\Omega$ $C = 22\ nF$

Therefore: $\tau = 10\ k\Omega \times 22\ nF = 220\ \mu s$
This value is suitable because it is much larger than the carrier period but smaller than the message period. The carrier period at 174 kHz is about 5.75 µs, while the message period at 1 kHz is 1 ms. Since 220 µs is between these two values, the capacitor does not discharge too much between carrier peaks, but it can still follow the 1 kHz message envelope.

Normally, a Germanium diode is preferred in an envelope detector because it has a lower forward voltage drop than a silicon diode. A lower forward voltage drop allows the diode to conduct more easily, which is useful when the signal amplitude is small. However, since a Germanium diode was not available

in the lab, a regular silicon diode was used. This may have reduced the detected output amplitude because a larger part of the input signal is lost across the diode.

In our measurement, the AM signal before detection had a carrier frequency of 174.67 kHz. After the detector, the recovered output signal had a frequency of 999.9 Hz and a peak-to-peak voltage of approximately 940 mVpp. This confirms that the detector successfully recovered the original 1 kHz message signal from the AM waveform.

Question 4. Why do you think the sound from the speaker is not that loud and what should we do to get louder sound?

The sound from the speaker was not very loud because the final output power delivered to the speaker was limited. Although the circuit produced a visible output signal on the oscilloscope, driving a speaker requires enough current as well as enough voltage. Op-amps such as the TL081 are not designed to deliver high output current to low-impedance loads like speakers. Therefore, even if the voltage gain is high, the speaker may not receive enough power to produce a loud sound.

Another reason is that the detected signal from the envelope detector was not very large. In our detector stage, the output was about 940 mVpp. This signal was then amplified by the final audio amplifier. At 100% modulation, the final output reached about 26.5 Vpp, and at 30% modulation it was around 8 Vpp. However, the actual sound level depends not only on peak-to-peak voltage but also on the load, current capability, speaker impedance, and coupling capacitor behavior.

The use of a silicon diode instead of a Germanium diode may also have reduced the detector output. Since a silicon diode has a larger forward voltage drop, part of the AM signal is lost before the capacitor charges. This lowers the recovered message signal amplitude, which can reduce the final sound level. The speaker sound can be made louder in several ways. First, a proper power amplifier stage can be added after the final op-amp stage. For example, an audio power amplifier IC or a transistor push-pull output stage could provide more current to the speaker. This would improve the power transfer to the speaker and increase the sound level.

Second, the modulation depth can be increased. In our experiment, the output was much larger at 100% modulation than at 30% modulation. A larger modulation depth creates a stronger envelope variation, which produces a larger detected audio signal. However, overmodulation should be avoided because it causes distortion.

Third, the detector efficiency can be improved by using a Germanium diode or a Schottky diode with a lower forward voltage drop. This would allow the detector to recover a stronger message signal.
Fourth, the output coupling capacitor and speaker impedance should be chosen properly. The coupling capacitor blocks DC but passes the AC audio signal. If the capacitor value is too small, low-frequency audio components may be attenuated. A larger coupling capacitor can improve low-frequency response and power transfer to the speaker.
In real electronic systems, the final output stage is very important. A signal may look strong on the oscilloscope, but it may still not be able to drive a speaker loudly if the circuit cannot provide enough current. This is why practical radios use dedicated audio power amplifier stages after demodulation.

Question 5. If you reverse the diode, will the receiver work? Why? Why not?

If the diode is reversed, the receiver will generally not work properly in the same way, because the envelope detector depends on the diode direction. The diode determines which peaks of the AM signal charge the capacitor. In the normal orientation, the diode conducts during the positive peaks of the AM signal and charges the capacitor so that the output follows the positive envelope.

If the diode is reversed, it will conduct during the negative peaks instead of the positive peaks. In that case, the detector may follow the negative envelope of the AM waveform. Theoretically, it may still recover an inverted version of the message signal if the rest of the circuit is referenced and connected correctly. However, in the practical receiver circuit used in this lab, reversing the diode would most likely prevent proper detection because the capacitor would charge in the opposite polarity and the following amplifier stage may not be biased correctly for that signal.

Also, the polarity of the detected signal matters because the detector output is connected to later stages through coupling and biasing components. If the capacitor charges in the opposite direction, the output DC level and waveform shape can become unsuitable for the audio amplifier. This can cause a very weak output, distorted output, or no audible sound at all.

Therefore, the diode should be connected in the correct direction for the expected envelope polarity. In our receiver, the correct diode orientation allowed the capacitor to charge at the AM peaks and discharge through the resistor, producing a recovered 1 kHz message signal. If the diode were reversed, the detector would no longer produce the same proper envelope output, and the receiver would not operate correctly as designed.

In real-life circuit design, diode polarity is very important in rectifier and detector circuits. A reversed diode can completely change the operating behavior of the circuit. This is why diode orientation must be checked carefully during breadboard implementation, especially in demodulators, power supplies, and protection circuits.

6. Conclusion

In this experiment, a complete AM radio receiver was designed, implemented, and tested by combining several circuit blocks studied throughout the course. The receiver included an LC tuner, cascaded op-amp amplification stages, an envelope detector, a final audio amplifier, and a speaker output. Each stage had a specific function in the system, and testing them step by step helped us understand how a complete communication circuit works in practice.

The results showed that the circuit operated successfully. The LC tuner passed the desired frequency component around 175 kHz, and the measured value of 175.78 kHz was very close to the theoretical resonant frequency. The amplifier stage increased the signal level significantly, although the experimental gain was lower than the theoretical value because of op-amp limitations, component tolerances, and breadboard effects. The detector stage successfully recovered the original message signal, giving an output frequency of approximately 999.9 Hz. Finally, the audio amplifier increased the detected signal enough to drive the speaker, and the full receiver produced an audible output.

The experiment also showed the importance of practical circuit limitations. In theory, each block can be analyzed using ideal equations, but the measured results are affected by real components, diode voltage

drops, gain-bandwidth limitations, parasitic capacitance, loading effects, and noise. Even with these differences, the measured results were consistent with the expected behavior of an AM receiver. The main objective of selecting, amplifying, demodulating, and listening to an AM signal was achieved.

Overall, this lab helped us understand how RLC circuits, op-amps, diodes, capacitors, and speakers can work together in a real electronic system. It also emphasized the importance of testing each circuit block separately before building the full system. The successful final output confirmed that the receiver design was valid and that the circuit performed its intended function effectively.